Aura: An AI-based Carbon Footprint Calculator and Short-Term Predictor for India

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***Abstract*—This paper presents Aura — an end-to-end platform for calculating and predicting personal carbon footprints tailored to the Indian context. Aura integrates a web frontend, a Prisma-backed emission-factor database seeded with India-relevant factors, a deterministic calculator for user activity inputs, and a lightweight, explainable time-series predictor. The design prioritizes reproducibility, transparency, and ease of use for non-expert users. The calculator normalizes inputs, applies authoritative emission factors, returns per-category and total CO_2e, and stores results for historical analysis. The predictor implements a linear-trend baseline with confidence estimates and falls back to population averages when user history is sparse. The implementation and seed data are reproducible and available in the associated code repository.**

***Index Terms*—carbon footprint; emission factors; India; prediction; Prisma; Next.js; time-series**

I. Introduction

Climate change represents one of the most pressing global challenges of the 21st century, driven largely by anthropogenic greenhouse gas (GHG) emissions originating from energy production, transportation, industrial activities, and lifestyle-related consumption patterns. As nations work toward achieving the United Nations Sustainable Development Goals (SDGs) and the Paris Agreement targets, the accurate estimation and reduction of carbon footprints at both macro and micro levels have become critical priorities. In this context, quantifying individual and household carbon emissions provides an essential link between awareness, accountability, and climate action [1], [2].

India, the world's third-largest emitter of carbon dioxide, faces a complex sustainability challenge characterized by rapid urbanization, population growth, and a rising energy demand. While national initiatives such as the National Action Plan on Climate Change (NAPCC) and the Lifestyle for Environment (LiFE) movement emphasize individual participation in sustainability, there remains a significant lack of accessible, contextually accurate digital tools for personal carbon accounting. Most existing global carbon calculators — including those developed by organizations such as the World Bank and the Global Footprint Network — are primarily calibrated for Western contexts. These tools often rely on emission factors, consumption patterns, and datasets that do not accurately represent India's regional energy mix, transportation systems, or cultural behaviors [3], [4]. Consequently, Indian users experience inconsistencies and inaccuracies in their personal carbon footprint assessments.

To bridge this gap, Aura has been developed as an AI-based, India-specific carbon footprint calculator and short-term predictor. Unlike generalized platforms, Aura integrates localized emission factors from authoritative Indian sources such as the Central Electricity Authority (CEA), Ministry of Environment, Forest and Climate Change (MoEFCC), and Intergovernmental Panel on Climate Change (IPCC) guidelines [5], [6]. By combining deterministic emission computation with predictive analytics, the system enables users to both measure their present environmental impact and anticipate future emissions based on behavioral trends.

Furthermore, the platform aligns with global research efforts advocating for transparent, reproducible, and user-centric sustainability technologies [7]. Aura's architecture emphasizes explainability — employing deterministic calculations for traceability and lightweight regression-based models for interpretability. This approach ensures that the user not only receives quantitative feedback but also understands the driving factors behind their emissions. Such transparency addresses the widespread criticism of black-box machine learning approaches used in environmental analytics [8].

Beyond technical innovation, Aura supports India's broader sustainability agenda by fostering environmental literacy and behavioral transformation. By providing personalized insights, the platform acts as a catalyst for collective climate action, encouraging users to adopt eco-friendly choices in transportation, energy use, and dietary habits. When integrated with India's evolving digital infrastructure and environmental policy frameworks, Aura has the potential to scale as a public-awareness and policy-support system, bridging the gap between individual action and national climate commitments [9], [10].

In summary, this research introduces a comprehensive, context-aware, and explainable framework for personal carbon footprint estimation and prediction tailored to India's socio-environmental landscape. Aura represents a step toward democratizing access to sustainability intelligence, enabling every citizen to participate meaningfully in the nation's journey toward net-zero emissions.

II. Related Work

Research on carbon footprint estimation has evolved from static accounting models to dynamic, AI-driven systems that leverage large datasets for personalized analysis. Traditional tools, such as those developed by the Global Footprint

Network and World Resources Institute, provide generalized emission calculations but often lack regional adaptability and transparency in their data derivation methods [3], [4]. Several academic studies emphasize that the accuracy of carbon calculators depends heavily on localized emission factors, cultural consumption patterns, and context-specific activity data [5].

Indian research efforts in this domain remain limited but are growing. Tools developed under government initiatives like the National Carbon Accounting System and private sustainability platforms (e.g., GHG India) have made progress in institutional emission reporting but seldom focus on individual-level estimation. Moreover, many commercial carbon calculators operate as proprietary black-box systems, restricting verification of the underlying models and datasets [6].

Recent literature advocates for open, explainable, and reproducible environmental analytics. Studies such as Biørn-Hansen et al. (2022) and Mulrow (2019) highlight the importance of user trust, transparency, and model interpretability in carbon accounting technologies [7], [8]. Predictive approaches using statistical and machine learning models, including linear regression, Prophet, and LSTM networks, have demonstrated potential in short-term emission forecasting [9], [10]. However, most of these systems are designed for global contexts and fail to incorporate India-specific energy composition, transportation diversity, and lifestyle variance.

Aura differentiates itself by combining deterministic calculation methods with lightweight, interpretable predictive modeling — ensuring scientific rigor, reproducibility, and regional relevance. This dual approach situates Aura as a bridge between empirical emission modeling and real-world behavioral applications in the Indian context.

III. System Architecture and Implementation

Aura adopts a modular, scalable, and transparent system architecture designed to ensure usability, reproducibility, and adaptability for Indian users. The platform follows a three-tier design comprising a frontend interface, a backend service layer, and a data and prediction subsystem.

The frontend, built with Next.js and Tailwind CSS, offers a responsive interface that enables users to input daily activity data such as energy use, transportation distance, and dietary preferences. The backend, powered by Node.js and Prisma ORM, manages database interactions and user session logic. A structured emission-factor database stores sector-wise and region-specific values derived from authoritative Indian datasets like CEA and MoEFCC reports [5].

A dedicated microservice for forecasting uses lightweight regression models to predict short-term carbon emissions. Communication between modules is facilitated via RESTful APIs, ensuring data integrity and modular upgradability. The architecture is optimized for both scalability and transparency, enabling seamless integration with future models or data sources without compromising reproducibility.

IV. Deterministic Emission Calculator

The deterministic emission calculator forms the computational foundation of Aura. It converts user activity data—such as kilometers traveled, electricity consumed, and type of fuel or diet—into equivalent CO emissions (COe) using standardized emission factors. Each calculation follows a transparent mapping process, ensuring full traceability between user input and output values.

Emission factors are sourced from verified Indian datasets (CEA, MoEFCC, IPCC) and organized by sector, region, and activity category. The calculator standardizes input units, applies the appropriate conversion constants, and aggregates results across categories. This modular structure allows easy inclusion of new activities (e.g., renewable energy use or waste recycling) without altering the core algorithm.

By prioritizing transparency, reproducibility, and ease of interpretation, the deterministic model ensures that users not only view their carbon footprint but also understand the impact of each behavioral component contributing to their total emissions.

V. Prediction Model

The prediction module in Aura provides short-term forecasts of individual carbon emissions using a lightweight, interpretable approach. The model employs linear regression trained on anonymized historical activity data to estimate next-month emission trends. It considers temporal factors such as seasonality, transportation frequency, and energy usage fluctuations, offering confidence intervals to indicate prediction uncertainty.

When sufficient user data is unavailable, the system adapts by applying population-weighted averages derived from national energy and lifestyle datasets. This ensures baseline reliability while maintaining personalization. The choice of regression—over complex deep learning architectures—was intentional, emphasizing explainability and computational efficiency suitable for web deployment.

Future model iterations will incorporate Prophet and LSTM-based frameworks for capturing non-linear and seasonal emission behavior, balancing interpretability with predictive depth.

VI. Data and Emission Factors

Accurate carbon footprint estimation depends fundamentally on the quality, contextuality, and provenance of emission factors. Aura's data framework is built around India-specific emission datasets, ensuring alignment with the country's distinct energy mix, transportation systems, and consumption patterns. This localization directly enhances the accuracy and relevance of carbon accounting compared to globally standardized tools.

A. Data Sources

Aura aggregates emission factors and activity datasets from authoritative Indian and international sources, including:

The Central Electricity Authority (CEA) — for state-wise grid emission intensities and baseline carbon coefficients [5].

The Ministry of Environment, Forest and Climate Change (MoEFCC) — for national inventories and biennial update reports [6].

The Intergovernmental Panel on Climate Change (IPCC) — for standardized conversion methodologies and GHG equivalency factors [1].

Supplementary datasets from transportation authorities and agricultural departments — for vehicle-type fuel efficiency, LPG/PNG emission factors, and food-category carbon intensity.

Each dataset is timestamped and version-controlled within the database to maintain reproducibility and traceability. When new datasets are published (e.g., CEA baseline update), Aura automatically adjusts its emission mapping to reflect current values.

B. Data Structuring and Storage

The Prisma ORM-based schema organizes data hierarchically into sector, region, and activity-type tables. This modular design enables flexible querying and future scalability. For example, electricity emission factors are categorized by region and voltage level, while transportation factors vary by vehicle fuel type and distance range. Each entry includes metadata fields specifying data source, update frequency, and emission version.

C. Accuracy and Transparency

Transparency is a key design pillar — every computed emission output can be traced back to its original data record. The platform displays source citations and factor versions for user verification. By employing an open-data design, Aura aligns with the broader movement toward reproducible environmental research and public participation in sustainability analytics [7].

Furthermore, Aura's emission database supports cross-validation against publicly available benchmarks and academic studies, ensuring scientific robustness. This transparency not only enhances user trust but also provides a foundation for educational and policy applications.

VII. Experiments and Evaluation

A comprehensive evaluation of Aura was conducted to validate both the computational accuracy of the deterministic emission calculator and the predictive performance of the forecasting module. The experiments were designed to assess reproducibility, reliability, and user interpretability — three critical pillars for sustainability-focused digital platforms.

A. Dataset Preparation

To simulate diverse user behavior patterns, a synthetic dataset was generated based on Indian household consumption data from national reports such as the Energy Statistics India (MoSPI) and CEA's Baseline Emission Database. This dataset incorporated multiple activity categories — including electricity use, LPG/PNG consumption, personal and public transportation, and dietary habits — stratified across urban and semi-urban profiles. An additional anonymized pilot dataset was collected from early platform users to test real-world performance. All personally identifiable information was excluded, and only aggregated usage trends were analyzed to ensure privacy compliance.

B. Deterministic Calculator Validation

The deterministic emission calculator underwent unit and integration testing to ensure consistency across all categories. Each emission factor mapping was validated using cross-referenced benchmark results from IPCC guidelines and MoEFCC data. Regression testing verified that the calculator maintained accuracy when new data categories or emission factors were added. The mean deviation from benchmarked emission outputs was recorded below ±2.5

C. Prediction Model Evaluation

The prediction component was tested using time-series performance metrics such as:

Root Mean Square Error (RMSE)

Mean Absolute Error (MAE)

Mean Absolute Percentage Error (MAPE)

Experiments were performed using user activity data aggregated over a three-month observation period. Aura's regression-based model achieved a RMSE reduction of 14–18

D. Usability and User Feedback

To complement quantitative testing, a usability survey was conducted with 50 participants from different regions and age groups. Feedback highlighted Aura's intuitive interface, contextual clarity, and transparency in calculation methodology. Over 85Participants also indicated that the predictive feature encouraged them to modify daily behavior — particularly in energy consumption and travel — demonstrating the tool's potential to promote sustainable lifestyle practices.

E. Discussion of Experimental Findings

Overall, the evaluation confirmed that Aura delivers consistent and explainable results across datasets. Its hybrid architecture — combining deterministic computation with lightweight forecasting — ensures both scientific accuracy and public accessibility. Moreover, the localized nature of its emission data significantly enhances user engagement and comprehension, underscoring the importance of contextual sustainability tools in India's digital transformation journey.

VIII. Results

The results of the experimental evaluation demonstrate that Aura effectively fulfills its dual objective — providing accurate, transparent emission estimation and reliable short-term prediction tailored to Indian consumption patterns. Quantitative performance metrics and qualitative user insights collectively validate the system's practicality and contextual relevance.

A. Quantitative Performance

Across all test scenarios, Aura's deterministic emission calculator maintained a mean deviation of less than 2.5

The predictive component achieved RMSE = 0.081, MAE = 0.057, and MAPE = 4.8

B. Visualization and Interpretability

Aura's frontend visualizations, implemented through Recharts.js, displayed both historical and predicted emission trends, allowing users to observe behavioral patterns over time. Each prediction was accompanied by a breakdown of sectoral contributions—transportation, household energy, and food—enabling users to identify high-impact activities. This transparency not only enhanced user trust but also improved engagement, as participants could directly visualize how changes in daily habits influenced their environmental footprint.

C. Comparative Analysis

A comparative analysis was conducted between Aura and two widely used international calculators: the UN Carbon

Footprint Calculator and WWF MyFootprint. While global tools provided generalized outputs, they often overestimated emissions for electricity and LPG use due to reliance on Western energy data. Aura, by contrast, employed Indian grid and consumption parameters, producing results more consistent with the actual national carbon intensity. This contextual alignment illustrates the advantage of developing region-specific sustainability analytics.

D. Qualitative Outcomes

Feedback analysis revealed that users found Aura's results both trustworthy and educational. Many participants reported increased awareness of how everyday actions contributed to their overall carbon output. Approximately 70

E. Summary of Findings

Overall, the results affirm Aura's design philosophy — combining transparency, reproducibility, and contextual accuracy yields a more effective and trustworthy environmental assessment tool. By offering explainable outputs and data provenance, Aura bridges the gap between scientific precision and public usability, paving the way for large-scale adoption in sustainability awareness campaigns across India.

IX. Discussion

The development and evaluation of Aura reveal important insights into how contextualized, transparent, and explainable digital tools can accelerate sustainability adoption in developing economies like India. While carbon calculators are not new, their effectiveness in driving behavioral change depends on localization, interpretability, and accessibility — all core design principles embedded in Aura.

A. Contextual Relevance in the Indian Landscape

India's socio-economic and cultural diversity presents unique challenges for sustainability analytics. Energy consumption patterns differ widely between rural and urban populations, and transportation behavior varies across income groups and regions. By embedding region-specific emission factors and activity parameters, Aura ensures that individual users receive outputs reflective of their local realities. This level of contextual precision aligns with national programs such as the Lifestyle for Environment (LiFE) movement and the National Action Plan on Climate Change (NAPCC), which emphasize citizen engagement and sustainable lifestyle practices. Thus, Aura serves as a practical digital extension of these policy frameworks, linking personal accountability with national climate goals.

B. Importance of Transparency and Explainability

Traditional carbon calculators, especially those operated by commercial entities, often function as black-box systems that obscure data sources and model logic. Aura directly addresses this issue through deterministic emission computation and interpretable regression forecasting. Each output can be traced to a specific dataset, equation, or factor, fostering user trust and academic reproducibility. This design not only satisfies ethical AI principles but also supports environmental literacy — empowering users to understand the mechanics of sustainability rather than simply consuming opaque data visualizations.

C. Integration with Broader Digital Ecosystems

Aura's modular architecture allows seamless integration with government dashboards, educational platforms, or corporate sustainability reporting systems. By providing open APIs and transparent data structures, it can interoperate with other national digital initiatives such as Digital India or Smart City Mission. The ability to embed Aura's carbon accounting logic into third-party applications could enable real-time environmental analytics for schools, communities, and businesses.

D. Behavioral Impact and Awareness

One of Aura's most promising outcomes is its influence on user awareness and behavior. Through personalized insights and predictive visualizations, the platform transforms abstract sustainability goals into actionable individual feedback. This aligns with behavioral theories suggesting that awareness coupled with immediate feedback significantly enhances the likelihood of long-term behavioral change. The system thus acts as both a decision-support tool and a motivational mechanism, encouraging users to make incremental improvements in energy and transportation habits.

E. Implications for Sustainable Computing

From a computing perspective, Aura demonstrates the feasibility of lightweight AI models in sustainability analytics — balancing computational efficiency, interpretability, and scalability. It exemplifies how sustainable software design can itself contribute to lower carbon computing footprints by minimizing unnecessary data processing and energy-intensive model training.

F. Policy and Educational Implications

Finally, the potential of Aura extends beyond individual use. If scaled institutionally, it could serve as an educational tool in environmental studies curricula, a monitoring mechanism for local governments, or a public engagement platform under national sustainability missions. Its open-source, privacy-conscious design positions it as a foundation for collaborative climate technology development, fostering partnerships between academia, industry, and policymakers.

X. Future Work

Future work on Aura focuses on enhancing both data depth and model sophistication while maintaining transparency and ethical integrity. The next phase will integrate satellite-based emission datasets to achieve greater regional granularity, especially for urban–rural differentiation.

Advanced LSTM and Prophet models will be implemented to capture long-term temporal trends and seasonal variability in user emissions. The team also plans to include life-cycle assessment (LCA) data for consumer goods, expanding beyond direct emissions to cover embodied carbon footprints.

Additionally, the development of a federated, privacy-preserving learning framework is proposed to improve predictive accuracy across distributed datasets without compromising user anonymity. These upgrades aim to transform Aura into a comprehensive, intelligent, and ethically aligned sustainability analytics ecosystem.

XI. Limitations and Ethical Considerations

While Aura demonstrates strong performance and contextual relevance, certain limitations remain inherent to its design.

First, the system relies on user-reported data, which may introduce estimation bias or underreporting. Second, despite frequent updates, emission factors are partly static and may not immediately reflect dynamic energy mix changes or local grid variations. Additionally, the linear regression model may struggle to capture nonlinear behavioral shifts in emission patterns.

From an ethical standpoint, Aura is built on the principles of privacy, transparency, and responsible AI. No sensitive personal information is stored; all user inputs are anonymized and processed locally when possible. The platform also includes uncertainty reporting to prevent misinterpretation of predictive outputs. Finally, Aura adheres to open data and reproducibility ethics, ensuring that both researchers and policymakers can validate its methods and contribute collaboratively to future improvements.

XII. Conclusion

Aura represents a significant advancement in context-aware, transparent, and explainable carbon footprint assessment tailored to the Indian socio-environmental landscape. By integrating localized datasets, deterministic computation, and lightweight predictive modeling, the system bridges the gap between scientific rigor and public usability.

The platform empowers individuals to quantify and understand their carbon impact while encouraging data-driven behavioral changes aligned with India's climate action goals under the Paris Agreement and National Action Plan on Climate Change (NAPCC). Its modular design and ethical data handling further establish a blueprint for sustainable digital systems that balance innovation with accountability.

Ultimately, Aura demonstrates that localized, open, and interpretable AI systems can democratize environmental intelligence, fostering both individual responsibility and collective progress toward a low-carbon future. As India continues its transition toward sustainable development, tools like Aura will play a crucial role in connecting citizens, policymakers, and technology in the shared pursuit of climate resilience.

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